

Calibration After Action Selection: Safety Claims for Offline RL under Moving Budgets

Anonymous submission

Abstract

Offline safe reinforcement learning methods often select deployment actions by thresholding learned cost or feasibility scores. Even when these scores admit marginal conformal calibration over candidate proposals, deployment-time action selection can concentrate calibration errors among the actions that are actually certified and executed. We study this calibration-after-selection problem for offline safe RL under moving budget queries. The intended contribution is not another constraint-adaptive policy optimizer, but a selection-aware auditor that targets false certification among issued action claims for a declared proposal stream, continuation policy, and support condition. The first required evidence is an exact counterexample showing that ordinary global or group conformal calibration can remain nominal over proposals while failing after action selection.

Introduction

The current draft route is maintained in `../idea_blueprint.md`. This file is a format-valid AAAI-27 skeleton only; paper-facing claims must not be added before the Phase 0 literature gate and Phase 1 exact selection-failure toy phenomenon are recorded.

Problem Setup

We consider an offline dataset collected under a behavior policy, a learned base policy, and deployment requests specified by cost budgets or safety constraints that may differ from those used during training or model selection.

Method

ACCS will be described as a selection-aware auditing protocol:

1. define the proposal stream, action-selection rule, continuation policy, fallback, and residual-cost target;
2. show why ordinary marginal conformal calibration can fail among selected claims;
3. control or diagnose false certification among issued claims using selection-aware calibration/risk control;

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4. report false certification, claim yield, risk-coverage, utility, fallback cost, and horizon-level audit metrics.

Experiments

The first experiment must establish a clean selection-induced calibration failure before broad benchmark runs. Direct competitors and reusable benchmark code are tracked in `../papers/literature_review.md`.

Conclusion

Placeholder.